

AI-ENABLED BUSINESS INTELLIGENCE

Week 2 — Data Foundations & Analysis Module

1. Overview of the Week

This week you build the two foundational skills of BI work: SQL querying and Python data analysis. You will complete 10 structured SQL practice queries, solve the famous SQL Murder Mystery challenge, and then load your dataset into Python, clean it, and produce 5 EDA outputs. By the end of this week, you can move data from a database into a notebook and start asking it meaningful questions.

2. What You Will Do This Week

- **Set up SQL environment:**
 - Install **DB Browser for SQLite** (sqlitebrowser.org — free, works on Windows/Mac/Linux) OR **MySQL Workbench** (dev.mysql.com/downloads/workbench/).
 - Download a sample dataset from Kaggle (e.g., Chinook, Sakila, or any dataset with multiple related tables) in CSV or SQLite format and load it into DB Browser.
- **Write 10 SQL Practice Queries** covering:
 - 2 basic SELECT + WHERE queries (filtering rows)
 - 1 ORDER BY query (sorting)
 - 2 GROUP BY + aggregate function queries (COUNT, SUM, AVG)
 - 1 HAVING query (filter after grouping)
 - 2 INNER JOIN queries (combining tables)
 - 1 LEFT JOIN query (preserving records from one side)
 - 1 subquery (a query inside another)
 - Save all 10 queries in a single .sql file with clear comments explaining each.
- **Solve the SQL Murder Mystery** (mystery.knightlab.com/walkthrough.html):
 - Complete the entire mystery by writing SQL queries.
 - Save your final solution query (the one that identifies the culprit) in your .sql file with a comment marking it clearly.
- **Python Data Analysis:**
 - Open a new Google Colab or Jupyter Notebook.
 - Load your chosen dataset (from Week 1) using `pd.read_csv()`.
 - **Clean the data:** handle missing values (drop or fill), fix incorrect data types (e.g., dates stored as strings), rename columns to lowercase with underscores, drop duplicate rows if any.

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- **Produce 5 EDA outputs:**

1. `df.shape` — dataset dimensions
 2. `df.describe()` — summary statistics for numerical columns
 3. `df['column'].value_counts()` — frequency of a categorical column
 4. Correlation matrix using `df.corr()` — with a brief interpretation
 5. 2 charts using matplotlib or seaborn (e.g., a histogram and a bar chart)
- Add markdown cells in your notebook explaining what each output shows.

3. Learning Outcomes for the Week

Week Outcome	Mapped to CLO
Write 10 structured SQL queries covering core SELECT, WHERE, JOIN, and GROUP BY operations	CLO-1
Solve a multi-step SQL problem independently (SQL Murder Mystery)	CLO-1
Load, clean, and explore a dataset end-to-end in pandas	CLO-2
Produce initial EDA visualisations using matplotlib and seaborn	CLO-2, CLO-4

4. Resources

4.1 YouTube Videos

#	Title / Content	Channel & Link
1	SQL Tutorial — Full Database Course for Beginners	freeCodeCamp — https://www.youtube.com/watch?v=HXV3zeQKqGY
2	SQL JOINS Explained	Alex The Analyst — https://www.youtube.com/watch?v=9yeOJ0ZMUYw
3	GROUP BY and HAVING in SQL	Alex The Analyst — https://www.youtube.com/watch?v=cQ7Iy5nO8Ks
4	SQL Murder Mystery Walkthrough	Alex The Analyst — https://www.youtube.com/watch?v=Z_6JdCjbFuE
5	Python pandas Tutorial — Full Course	Corey Schafer — https://www.youtube.com/playlist?list=PL-osiE80TeTsWmV9i9c58mdDCSskIFdDS

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6	Exploratory Data Analysis in pandas	Rob Mulla — https://www.youtube.com/watch?v=xi0vhXFPegw
7	Data Cleaning with pandas	Keith Galli — https://www.youtube.com/watch?v=bDhvCp3_IYw
8	Jupyter Notebook Tutorial	Corey Schafer — https://www.youtube.com/watch?v=HW29067qVWk
9	matplotlib Tutorial for Beginners	Corey Schafer — https://www.youtube.com/watch?v=UO98IJQ3QGI
10	SQL Subqueries Explained	Alex The Analyst — https://www.youtube.com/watch?v=GpC0XyF9hag

4.2 Blogs and Articles

#	Title	Source & Link
1	SQL Tutorial — W3Schools	W3Schools — https://www.w3schools.com/sql
2	Mode SQL Tutorial	Mode Analytics — https://mode.com/sql-tutorial
3	SQL Murder Mystery	Knight Lab — https://mystery.knightlab.com
4	pandas Getting Started	pandas Docs — https://pandas.pydata.org/docs/getting_started
5	Data Cleaning with pandas — Practical Guide	Real Python — https://realpython.com/python-data-cleaning-numpy-pandas
6	Exploratory Data Analysis Checklist	Towards Data Science — https://towardsdatascience.com
7	Handling Missing Data in pandas	pandas Docs — https://pandas.pydata.org/docs/user_guide/missing_data.html
8	SQL JOINS — Visual Guide	CodeProject — https://www.codeproject.com/Articles/33052/Visual-Representation-of-SQL-Joins
9	matplotlib Documentation	matplotlib — https://matplotlib.org/stable/users/index.html
10	DB Browser for SQLite — How to Use	DB Browser Docs — https://sqlitebrowser.org

4.3 MOOCs / Courses

#	Course	Platform & Link
1	SQL for Data Science	Coursera (UC Davis) — https://www.coursera.org/learn/sql-for-data-science

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2	Data Analysis with Python	freeCodeCamp (Free) — https://www.freecodecamp.org/learn/data-analysis-with-python
3	Python + SQL for Data Science	IBM Data Science Professional Certificate (Coursera) — https://www.coursera.org/professional-certificates/ibm-data-science
4	SQL Basics	Khan Academy (Free) — https://www.khanacademy.org/computing/computer-programming/sql

5. Deliverables

Deliverable 1: SQL Query File

- A single .sql file containing:
 - 10 practice queries (each preceded by a comment like -- Query 1: List all customers from Germany).
 - The SQL Murder Mystery final solution query (clearly marked with -- SQL Murder Mystery Solution).
- Each query must actually run and return results in DB Browser or MySQL Workbench.
- File name: W2_SQLQueries_[InternID].sql

Deliverable 2: Jupyter / Colab Notebook (.ipynb)

- A single notebook file (.ipynb) or a Google Colab shareable link.
- Must include:
 - Loading the dataset with `pd.read_csv()`.
 - Cleaning steps (minimum 3 cleaning operations).
 - 5 EDA outputs as specified above (shape, describe, value_counts, correlation matrix, 2 charts).
 - Markdown cells explaining what each output reveals.
- Cell outputs must be visible (run all cells before submitting).
- File name: W2_EDA_Notebook_[InternID].ipynb (or Google Colab link with sharing set to "Anyone with link can view").

6. Submission Instructions

- Submit W2_SQLQueries_[InternID].sql and W2_EDA_Notebook_[InternID].ipynb via the **TBI-GEU LMS Assignment box for Week 2** by **Sunday 11:59 PM IST**.
- If using Google Colab, share the link (with "Anyone with link can view") in the TBI-GEU LMS text/link field.
- Submit the **Weekly Progress Report** via the TBI-GEU LMS Form by **Sunday 11:59 PM IST**.

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Week 3 — Data Visualisation

1. Overview of the Week

This week you move from "producing charts" to "choosing the right chart". Each chart type exists to answer a specific kind of question — comparisons, trends, distributions, relationships, compositions. You will build 5 different chart types in Python, document why you chose each for the question it answers, and export your 3 best charts as publication-quality PNGs with written narratives.

2. What You Will Do This Week

- **Build 5 chart types in Python** using matplotlib and/or seaborn, on your dataset from Week 1–2:
 1. **Line chart** — for trends over time (e.g., monthly sales).
 2. **Bar chart** — for comparing categories (e.g., revenue by region).
 3. **Pie chart** — for composition or share (e.g., market share by brand). Use sparingly — pie charts only work with ≤ 5 categories.
 4. **Scatter plot** — for relationships between two numeric variables (e.g., price vs sales volume).
 5. **Heatmap** — for correlation between multiple variables or category $\times$ category density (e.g., correlation matrix heatmap or day $\times$ hour booking counts).
- **Polish every chart:** each chart must have a descriptive title, axis labels with units, a colour palette choice (not the default matplotlib blue everywhere), and a legend if there are multiple series.
- **Document chart rationale:** for each of the 5 charts, write 1–2 sentences on which business question the chart answers and why that chart type is the right choice.
- **Apply chart selection principles:** refer to the chart chooser guide from the Financial Times Visual Vocabulary (ft-interactive.github.io/visual-vocabulary) or Cole Nussbaumer Knaflic's guides.
- **Select your 3 best charts** — these should be the ones most directly relevant to your business question.
- **Export 3 final charts as publication-quality PNGs:**
 - Use `plt.savefig('chart1.png', dpi=300, bbox_inches='tight')` for high resolution.
 - Minimum size: 1200×800 px.
- **Write a Visual Narrative for each of the 3 exported charts:**
 - 150 words each.
 - Structure: (a) what the chart shows, (b) one specific observation or finding, (c) what this means for the business decision.

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3. Learning Outcomes for the Week

Week Outcome	Mapped to CLO
Build 5 chart types in Python with proper titles, labels, and colour palettes	CLO-2, CLO-4
Apply chart-selection principles to match chart type to business question	CLO-4
Export publication-quality charts as high-resolution PNGs	CLO-4
Write concise visual narratives that translate charts into business language	CLO-4, CLO-7

4. Resources

4.1 YouTube Videos

#	Title / Content	Channel & Link
1	matplotlib Full Tutorial	Corey Schafer — https://www.youtube.com/playlist?list=PL-osiE80TeTvipOqomVEeZ1HRrcEvtZB
2	seaborn Tutorial for Beginners	Derek Banas — https://www.youtube.com/watch?v=6GUZXDef2U0
3	Data Visualisation in Python — Full Course	freeCodeCamp — https://www.youtube.com/watch?v=a9UrKTVEeZA
4	How to Choose the Right Chart Type	Storytelling with Data — https://www.youtube.com/watch?v=IHgnM80i8sM
5	5 Types of Charts Every Analyst Should Know	Alex The Analyst — https://www.youtube.com/watch?v=xs1Q4qTrmjM
6	Seaborn Heatmap Tutorial	StatQuest with Josh Starmer — https://www.youtube.com/watch?v=oMtDyOn2TCc
7	Scatter Plot Explained	StatQuest with Josh Starmer — https://www.youtube.com/watch?v=LZzqlzSL1bs
8	Chart Selection Guide	Storytelling with Data — https://www.youtube.com/watch?v=8EMW7io4rSI
9	Data Visualisation Best Practices	Google Analytics — https://www.youtube.com/watch?v=9x83Ayo_f8E
10	Python Charts That Tell a Story	Ken Jee — https://www.youtube.com/watch?v=9nBkEudUaqq

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4.2 Blogs and Articles

#	Title	Source & Link
1	matplotlib Tutorial — Official	matplotlib Docs — https://matplotlib.org/stable/tutorials/index.html
2	seaborn Documentation	seaborn — https://seaborn.pydata.org/tutorial.html
3	Data Visualisation Chart Selection	Data Visualisation Catalogue — https://datavizcatalogue.com
4	FT Visual Vocabulary	Financial Times — https://ft-interactive.github.io/visual-vocabulary
5	Storytelling with Data Blog	Cole Nussbaumer — https://www.storytellingwithdata.com/blog
6	Chart Selection Guide	Chartio (Atlassian) — https://chartio.com/learn/charts
7	Colour Theory in Data Visualisation	Datawrapper Blog — https://blog.datawrapper.de
8	Visualisation Best Practices	Google Data Studio Help — https://support.google.com/looker-studio
9	How to Make Good Charts	HBR — https://hbr.org/2016/06/visualizations-that-really-work
10	Seaborn vs Matplotlib — When to Use Which	Real Python — https://realpython.com/python-matplotlib-guide

4.3 MOOCs / Courses

#	Course	Platform & Link
1	Data Visualisation with Python	Coursera (IBM) — https://www.coursera.org/learn/python-for-data-visualization
2	Data Visualisation and Communication	Coursera (Duke) — https://www.coursera.org/learn/analytics-mba
3	Data Analysis with Python (Visualisation Module)	freeCodeCamp (Free) — https://www.freecodecamp.org/learn/data-analysis-with-python
4	Fundamentals of Visualisation with Tableau	Coursera (UC Davis) — https://www.coursera.org/learn/dataviz-fundamentals

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5. Deliverables

Deliverable 1: Python Notebook with All 5 Charts

- Single .ipynb file or Google Colab link.
- Must contain 5 separate code cells, one for each chart type (line, bar, pie, scatter, heatmap).
- Each chart cell followed by a markdown cell explaining: the business question the chart answers, why this chart type was chosen.
- All 5 charts must actually render when cells are run.
- File name: W3_Visualisations_[InternID].ipynb (or Colab link).

Deliverable 2: 3 Annotated PNG Charts

- 3 PNG files exported from your Python notebook at minimum 300 DPI.
- Each chart must have a descriptive title, labelled axes with units, clean colour palette, and legend if applicable.
- File names: W3_Chart1_[InternID].png, W3_Chart2_[InternID].png, W3_Chart3_[InternID].png.

Deliverable 3: Visual Narratives PDF

- Single PDF combining all 3 charts + narratives.
- For each chart (1 chart per page):
 - The PNG embedded in the PDF.
 - A 150-word Visual Narrative below it (± 10 words).
 - Narrative structure: what the chart shows (50 words), one specific observation (50 words), what this means for the business decision (50 words).
- File name: W3_VisualNarratives_[InternID].pdf

6. Submission Instructions

- Submit W3_Visualisations_[InternID].ipynb (or Colab link), all 3 PNG files, and W3_VisualNarratives_[InternID].pdf via the **TBI-GEU LMS Assignment box for Week 3** by **Sunday 11:59 PM IST**.
- Submit the **Weekly Progress Report** via the TBI-GEU LMS Form by **Sunday 11:59 PM IST**.
- Post on the **Cohort Forum** — share one technical or analytical challenge you faced this week (mandatory Week 3 Forum contribution).